

Since a real-valued bivariate function $z = f(x, y)$ has more than one partial derivative, it would be convenient to keep them all in one place for when we need one or more of them. The following definition gives us this convenient location.

Example. Suppose that $z = f(x, y)$ is a real-valued bivariate function. The **gradient of f** , denoted ∇f , is the vector-function whose components are the corresponding partial derivatives of f :

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle.$$

Now, let's reconsider $z = f(x, y) = \sqrt{1 - x^2 - y^2}$ restricted to the octant where $x, y, z \geq 0$. We consider an arbitrary point $P(x_0, y_0)$ in D_f and look at the x - and y -slices at P .

In this discussion, the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ were returning information about the rate at which f changes in the x - and y -direction, respectively. Alternatively, we think of this in the $\mathbf{i} = \langle 1, 0, 0 \rangle$ and $\mathbf{j} = \langle 0, 1, 0 \rangle$ directions.

In general, from the point $f(x_0, y_0)$, we are free to 'walk' in any direction. We can pick any directional unit vector $\mathbf{u}$ and ask how f changes in *that* direction. This object is defined as follows:

Definition. Suppose that $z = f(x, y)$ is a real-valued bivariate function. Then the *directional derivative of f in the direction of the unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$* and is given by

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u} = f_x(x, y)u_1 + f_y(x, y)u_2.$$

Example. Find the derivative of $f(x, y) = xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.

Revisiting the directional derivative, we can reformulate it as the following:

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u} = \|\nabla f\| \|\mathbf{u}\| \cos(\theta) = \|\nabla f\| \cos(\theta),$$

where θ is the angle between the vectors $\mathbf{u}$ and ∇f . We can make some observations:

- (i) Observe that f is increasing the fastest when $D_{\mathbf{u}}f(x, y) = \|\nabla f\| \cos(\theta)$ is maximized. This occurs when $\theta = 0$. This means f is increasing most rapidly when $\mathbf{u}$ *points in the direction of* ∇f . Specifically, when $\mathbf{u} = \frac{\nabla f}{\|\nabla f\|}$. Moreover, this maximal value is given by $\|\nabla f\|$.
- (ii) Similarly, f decreases most rapidly when $\mathbf{u}$ points in the direction of $-\nabla f$. Specifically, when $\mathbf{u} = -\frac{\nabla f}{\|\nabla f\|}$. Moreover, this minimal value is given by $-\|\nabla f\|$.
- (iii) If $\mathbf{u}$ is orthogonal to ∇f , then $D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u} = 0$ and there is no change in f .

Example. Consider the function $f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$

- (i) Draw a picture of this surface and locate the point at $(1, 1)$.

- (ii) Find the direction in which f increases most rapidly at $(1, 1)$ and the direction in which f decreases most rapidly at $(1, 1)$.

- (iii) Find the direction for which f exhibits no change at $(1, 1)$.

Now, suppose we are looking at a 2-space curve given by $\mathbf{r}(t) = \langle g(t), h(t) \rangle$:

Suppose further that this curve is actually a level curve for a surface generated by a real-valued bivariate function $z = f(x, y)$. Then we would have $f(g(t), h(t)) = c$ for some constant c . Differentiating both sides of this equation with respect to t yields:

$$\frac{d}{dt}f(g(t), h(t)) = \frac{d}{dt}c \implies$$

Therefore ∇f is orthogonal to the tangent vector $\mathbf{r}'(t)$ and hence the gradient is always normal to the level curve!

Observe, then, that a convenient way to construct the equation of a tangent line to a surface's level curve at (x_0, y_0) is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0.$$

Example. Consider the function $f(x, y) = \frac{1}{4}x^2 + y^2$. Find an equation for the line tangent to the level curve of f at $(-2, 1)$.

Thus, if we can do the above on a 2-space level curve to yield equations for the 1-dimensional tangent line, it makes sense that we can do the same process on a 3-space level surface to yield equations for the 2-dimensional tangent planes. We will see this in the next section.

Lastly, we take up the issue of having never developed a ‘chain rule’ for our conversations involving particles traveling along paths through space. If $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ is a smooth path C through the xy -plane and $z = f(\mathbf{r}) = f(x(t), y(t))$ is some scalar function evaluated along C , then we know that

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

But now we can rewrite this as:

$$\frac{d}{dt}f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Thus, with different notation of ‘derivative’, ‘functions’, ‘inside’, ‘outside’, and ‘multiplication’, we can still recover a chain rule for these objects.